Ryan Gupta

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Education

University of Michigan Expected May 2028

B.S. Computer Science and Robotics GPA: 3.4/4.0

Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Operating Systems (in progress), Probability & Statistics, Linear Algebra, Controls, Cybersecurity, Digital Logic / Verilog

Experience

Visa May 2026 – Aug 2026

Software Engineer Intern *Austin, TX*

- Built a database-backed deduplication system for a multi-tenant monitoring platform, eliminating repeated alerts and redundant LLM triage for identical production failures.
- Built an autonomous incident-response service in Java and Spring Boot using Claude, ClickHouse logs, and operational runbooks to diagnose failures and automatically execute remediation or escalation paths.
- Rebuilt Rubrik deployment and re-enrollment automation across 9 production servers, eliminating 7 observed backup failures and reducing provisioning time from **4 hours** to under **5 minutes**.

Michigan Advanced Robotics Competition (MARC) Mar 2025 – Present

Co-Founder & AI/ML Lead *Ann Arbor, MI*

- Co-founded a University of Michigan competition robotics team that generated **250+ student interest signups**, helping establish the organization, recruit its first engineering cohort, and launch technical development.
- Architecting a C-based embedded control stack using CAN-connected DJI M3508 motors and Jetson compute, with emphasis on deterministic execution, hardware-software interfaces, and real-time control.

University of Michigan Formula SAE Sep 2024 – Apr 2026

Software Engineer *Ann Arbor, MI*

- Cut ROS 2 LiDAR processing latency by 50% and missed obstacles by 42% by profiling C++ and Python nodes, optimizing clustering algorithms, and refining frame-association logic.
- Used test-vehicle telemetry and repeated experiments to validate perception changes end to end before integration into the broader autonomous vehicle software stack.

University of Waterloo Alternative Fuels Team (UWAFT) May 2025 – Aug 2025

Software Engineer *Waterloo, ON*

- Built repeatable MATLAB/Simulink experiments for lane-centering control, enabling systematic comparison of controller settings and vehicle-response metrics across runs.
- Standardized simulation and logging workflows in MATLAB, improving reproducibility and making controller experiments easier to compare across team members.

Skills

Programming Languages: C++, C, Python, Java, TypeScript, SQL, MATLAB, Verilog

Frameworks & Tools: Linux, ROS 2, Spring Boot, FastAPI, React, OpenCV, MATLAB/Simulink, FAISS, ClickHouse, SQLite, Docker, AWS, Git, CAN Bus, Jetson

Projects

FinSearch Aug 2026

- Built an end-to-end financial search product over **36,599** SEC filing passages using Python, FastAPI, FAISS, SQLite, semantic embeddings, and hybrid retrieval.
- Built an evaluation harness comparing lexical, semantic, and hybrid search quality and latency, measuring up to **0.893 NDCG@10** and sub-30 ms p95 hybrid retrieval.

Code Duel Apr 2026

- Implemented a **100 ms** server-tick multiplayer engine with Socket.io, maintaining synchronized session state and deterministic update cadence across connected clients.
- Implemented isolated code execution for user-submitted JavaScript and Python and connected it to matchmaking, session state, and real-time gameplay.